Dresden University of Technology

Computational Logic ∴ Group

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Advanced Problem Solving and Search

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Exercises 3

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Problem 1.
Consider the following **job-shop scheduling problem** with 3 jobs and 3 machines. Each job must be processed on the machines in a fixed order; each operation requires exclusive use of its machine; and no job can start its next operation before the previous one finishes. The goal is to minimise the **makespan** (total completion time).

	Operation 1	**Operation 2**	**Operation 3**
Job 1	M1 (dur. 3)	M2 (dur. 2)	M3 (dur. 2)
Job 2	M2 (dur. 2)	M1 (dur. 3)	M3 (dur. 1)
Job 3	M1 (dur. 2)	M3 (dur. 3)	M2 (dur. 1)

Design a Tabu Search algorithm for this problem by answering the following questions.

1. **Representation.** How would you represent a solution? Discuss any design choices and their consequences for feasibility.
2. **Evaluation function.** Define the evaluation function. How does minimisation fit the standard Tabu Search framework (which maximises)?
3. **Initial solution.** How do you generate a feasible starting schedule? What is the makespan of the natural ordering (Job 1, Job 2, Job 3) on every machine?
4. **Neighbourhood and tabu list.** Define the neighbourhood move and explain what goes on the tabu list. What is a sensible tabu tenure and why?
5. **Aspiration criterion.** Describe an aspiration criterion suitable for this problem and give a concrete example of when it should trigger.
6. **Feasibility.** Some swaps in the neighbourhood create circular dependencies (deadlocks). How should the algorithm handle such infeasible neighbours?

Problem 2.
Consider a SAT problem with 6 Boolean variables $x_1, \ldots, x_6$ and the following **weighted** CNF formula. The evaluation function is the **weighted sum of satisfied clauses** (higher is better).

Clause	Formula	Weight
C_1	x_1	3
C_2	x_2	3
C_3	$\neg x_1 \vee x_3$	2
C_4	$\neg x_2 \vee x_4$	2
C_5	$x_1 \vee \neg x_3 \vee x_5$	1
C_6	$\neg x_4 \vee \neg x_5 \vee x_6$	1
C_7	$\neg x_1 \vee \neg x_2 \vee \neg x_6$	1
C_8	$x_3 \vee x_4 \vee \neg x_6$	1

The neighbourhood of an assignment is all single-variable flips. The tabu list uses **recency-based memory** (variant 1: $M(i)$ stores the iteration at which x_i was most recently flipped; a flip is tabu if $t - M(i) \leq$ tenure). When all non-tabu moves lead to a worse solution, **frequency-based memory** H is used: the adjusted score for flipping x_i is $\mathit{eval}(\mathbf{x}') - 0.7 \cdot H(i)$, where $H(i)$ counts how many times x_i was flipped in the last h iterations.

Parameters: tenure $= 4$, penalty coefficient $= 0.7$.

State at the start of iteration $t = 7$:

- Current assignment: $\mathbf{x} = (1, 0, 1, 1, 0, 1)$, $\mathit{eval}(\mathbf{x}) = 11$
- Recency memory: $M = [1, 5, 4, 3, 2, 6]$ ($M(i)$ = iteration when x_i was most recently flipped)
- Frequency memory: $H = [2, 4, 6, 3, 2, 5]$ (counts over last h iterations)
- Best solution so far: $\mathit{best} = 12$

Execute **three iterations** of Tabu Search ($t = 7, 8, 9$), showing your work at each step. For each iteration:

(a) Determine which variables are currently tabu.

(b) Evaluate all non-tabu flips. Is there an improving move?

(c) If no non-tabu flip improves, check the aspiration criterion. Does it apply?

(d) If neither applies, use frequency-based memory to choose.

(e) State the move made, update M and *best* as needed, and increment t.

Hint: each iteration exercises a different mechanism — work through them in order and think about which mechanism applies at each step.